

Feature	Description	Value	Unit
Overall Length	Total auger length	444	mm
Flighted Length	Effective helical section	286	mm
Number of Flights	Total helical turns	4.5	rev
Pitch	Distance between flights	60	mm
Outer Diameter	Maximum auger diameter	60	mm
Core Shaft Diameter	Shaft diameter under flight	16	mm
Upper Smooth Shaft Length	Non-flighted upper section	110	mm
Lower Clearance Length	End clearance before thread	16	mm
Threaded Length	Thread for rotary plate	35	mm
Thread Diameter	Thread nominal diameter	14	mm
Flight Thickness	Helical ribbon thickness	4	mm
Radial Flight Height	Flight radial height	22	mm
Radial Clearance	Clearance to tube wall	1.0–1.5	mm
Fillet	Flight leading edge	1×45°	-
Fillet Radius	Flight to shaft root		4 mm
Fillet Radius	Fillet between Main shaft and smooth shaft		9 mm
Fillet Radius	The end of the pin attached to the smooth shaft	R1 = 3.5	mm
Fillet Radius	The point of contact between the pin and the post		0.5 mm
Fillet Radius	The end of the pin attached to the smooth shaft	R2 = 3.49	mm
Fillet Radius	End of the column connected to the threaded column		2 mm
Chamfer	Chamfer end of smooth shaft	1×45°	
Chamfer	Start of the thread column from the bottom	0.5×45°	
Fillet radius	Smoothing the beginning and end of the flight		0.5 mm

Notes

Bottom tip to top smooth end

Active dosing zone

Single start

Constant pitch

Flight OD

Main core

Connection to auger shaft

Between flight and thread

M14×1.5

Metric fine

Axial thickness

$(OD - Core) / 2$

Avoid rubbing

Smooth flow

Stress relief

On the right side of the smooth column

Contact with the smooth column

On the left side of the smooth column

Features	Value	Unit
Out Diameter of spinner plate		100 mm
Cut-Extrude diameter in the middle		14.5 mm
Diameter of the flat plate before the edge		69.55 mm
Thickness of the spinner plate (shell)		3 mm
Angle of inclination of the dish's edge		55 °
Chamfer in the edge of Cut-Extrude	0.5×45	°
Fillet radius		3 mm
Cut and extrude inwards by the amount (shell)		10 mm
Shell		3 mm

Description (English)

Total outer diameter of the spinner plate, defining the maximum extent of the inclined edge.

Central bore diameter, sized to allow passage of the auger shaft or drive spindle.

Diameter of the flat central region before the start of the sloped edge; material rests here before sliding outward.

Wall thickness of the spun or formed shell that constitutes the plate.

Conical angle of the outer rim; governs the radial throw and flow pattern of discharged material.

Small 45° chamfer on the bore edge to ease assembly and reduce stress concentration.

Blend radius between adjoining surfaces to minimize stress risers and improve manufacturability.

Axial depth of the dished shell, measured from the bottom plane to the top edge of the inclined rim.

Indicates the part is a thin-walled shell (constant 3 mm thickness) rather than a solid disc.

Description

القطر الخارجي الكلي للصحن الدوار، وهو القطر الأقصى للصحن الذي يحد من حافة الإمالة الخارجية. قطر الفتحة المركزية المخزومة في منتصف الصحن، تُستخدم لتمرير العمود أو المغزل الذي يُثبت عليه اللولب. قطر المنطقة المستوية من الصحن قبل أن تبدأ الحافة المائلة، وهي المنطقة التي تستقيم عليها المادة قبل أن تنزلق نحو الحافة. سمك الصحن الدوار نفسه، أي سمك الجزء المعدني المكون للصحن. زاوية ميل الحافة الخارجية للصحن، تُستخدم لتوجيه تدفق المادة نحو الخارج أثناء الدوران. شطبة صغيرة عند حافة الفتحة المركزية، تُستخدم لتسهيل التجميع وتقليل التركيز الإجهادي. نصف قطر التنعيم عند التقاء الأسطح، لتقليل التركيز الإجهادي وتسهيل التصنيع. الارتفاع الرأسي للصحن من السطح السفلي إلى أعلى حافته، أي العمق الداخلي للصحن. يشير إلى أن الصحن مصمم كـ"قشرة" (shell) أي جسم مجوف ذو سماكة ثابتة من 3 مم.

Feature	Value	Unit
Overall Length	130	mm
Outer diameter	30	mm
Length of cut-Extrude on the upper	30	mm
Diameter of cut-Extrude on the upper	15	mm
Length of cut-Extrude on the down	58	mm
Diameter of cut-Extrude on the down	20	mm
Vertical cut-Extrude length of J-slot	40	mm
Diameter of the J-slot	7	mm
Horizontal cut-Extrude length of J-slot	13	mm
Vertical Length of the cut on the upper surface	17	mm
Horizontal Length of the cut on the upper surface	25	mm
The diameter of the circle allows the cross pin to fit.	12	mm
Distance of the circle from the top surface of the column	15	mm
Fillet Radius	0.5	mm

Description

Total auger shaft length

Maximum auger shaft diameter

Cutting and extruding from the top

The diameter of the cut at the top is intended for the entry of the shaft that connects to the gearbox.

Cutting length - extrusion from the bottom at the point of contact with the smooth shaft of the screw containing t

The diameter of the cut from below is for the smooth shaft of the screw to enter.

Total length of J-slot cutting and extrusion

The J-slot design allows the insertion of the two pins located in the smooth shaft.

The length of the horizontal cut from the end of the vertical cut of the J-slot

The Vertical length of the cut is on the upper surface of the drill bit on both faces of the bit in order to create a cir

The Horizontal length of the cut is on the upper surface of the drill bit on both faces of the bit in order to create a

The diameter of the circle at the top, on the cut surface of the shaft, allows for the insertion of a cross-pin to secu

The distance of the circle from the top surface of the column, and this circle allows the entry of the contact pin cal

Smoothing of all edges

No.	Force Type	Equation	Symbols Meaning
1	Material Weight Force	$F_w = m \cdot g$	m = mass (kg), $g = 9.81 \text{ m/s}^2$
2	Axial Force (Helix)	$F_{ax} = m \cdot g \cdot \sin(\alpha)$	α = helix angle
3	Normal Force	$F_n = m \cdot g \cdot \cos(\alpha)$	—
4	Friction Force	$F_f = \mu \cdot F_n$	μ = friction coefficient
5	Axial Friction Force	$F_{ax,f} = \mu \cdot m \cdot g \cdot \cos(\alpha)$	—
6	Bulk Pressure Force	$F_p = \sigma \cdot A$	σ = bulk stress
7	Shear Force	$F_s = \tau \cdot A$	τ = shear stress
8	Rotary Plate Load	$F_{plate} = (m_p + m_m) \cdot g$	plate + material mass
9	Radial Force	$F_r = F_n \cdot \sin(\alpha)$	—
10	Inertial Force	$F_i = m \cdot a$	a = acceleration
11	Centrifugal Force	$F_c = m \cdot \omega^2 \cdot r$	ω = angular speed
12	Shaft Tangential Force	$F_t = T / r$	T = torque
13	Key Shear Force	$F_{key} = T / r_{key}$	r_{key} = key radius
14	Cross Pin Force	$F_{pin} = T / r_{pin}$	r_{pin} = pin radius
15	Equivalent Static Force	$F_{eq} = T / r_m$	r_m = mean radius

Engineering Use
Gravity load of material
Material conveying force
Used for friction
Material resistance
Affects torque
High fill cases
Cohesion resistance
Load on auger
Bending analysis
Startup condition
Rotating elements
Shaft stress
Key safety
Pin safety
FEA loading

Force Type	Equation	Symbolic Form	Input Parameter	Calculated Result	Units
Weight Force	$F = m \cdot g$		4.905 m, g	4.905 N	
Axial Friction Force	$F = \mu \cdot m \cdot g$		1.962 μ , m, g	1.962 N	
Torque from Weight	$T = F \cdot r$		0.14715 m, g, r	0.14715 N·m	
Centrifugal Force	$F = m \cdot r \cdot \omega^2$		6 m, r, ω	6 N	

Notes

Parameters

Mass (kg)	0.5
Gravity (m/s ²)	9.81
Friction Coefficient	0.4
Radius (m)	0.03
Angular Speed (	20

Gravitational force of material

Axial resistance

Torque due to mass

Rotational force

Parameter / Equation	Symbol	Value	Unit	Formula (Excel-driven)
Outer Diameter	D	0.06	m	
Shaft Diameter	Ds	0.016	m	
Pitch	p	0.06	m	
Number of Turns	n	4.5	-	
Fill Efficiency	η_v	0.6	-	
Bulk Density	ρ_b	1300	kg/m ³	
Cross-sectional Area	A		m ²	#VALUE!
Volume per Pitch	Vp		m ³	#VALUE!
Effective Volume per Pitch	Veff		m ³	#VALUE!
Total Volume	Vt		m ³	#VALUE!
Mass per Dose	m		kg	#VALUE!

Description

Auger outer diameter

Auger shaft diameter

Auger pitch

Effective auger turns

Volumetric fill efficiency

Bulk density of salt

Annular cross-sectional area

Volume conveyed per pitch

Effective volume considering fill efficiency

Total conveyed volume

Mass delivered per dose

No.	Calculation Item	Equation	Description (Arabic / English)
1	Annular Area	$A = \pi/4 (D^2 - D_s^2)$	Effective annular area / مساحة المقطع الحلقي
2	Volume per Revolution	$V_{rev} = A \cdot p$	Volume per revolution / الحجم لكل لفة
3	Effective Volume	$V_{eff} = \eta_v \cdot V_{rev}$	Effective volume / الحجم الفعلي
4	Total Auger Volume	$V_{total} = V_{eff} \cdot n$	Total volume / الحجم الكلي
5	Volumetric Flow Rate	$Q_v = V_{eff} \cdot N$	Flow rate / معدل التدفق الحجمي
6	Mass per Revolution	$m_{rev} = \rho_b \cdot V_{eff}$	Mass per rev / كتلة لكل لفة
7	Mass Flow Rate	$\dot{m} = \rho_b \cdot Q_v$	Mass flow / معدل التدفق الكتلي
8	Fill Time	$t = m_{dose} / \dot{m}$	Fill time / زمن التعبئة
9	Revolutions per Dose	$N_{dose} = m_{dose} / m_{rev}$	Required turns / عدد اللفات
10	Rotation Angle	$\theta = 360 \cdot N_{dose}$	Rotation angle / زاوية الدوران
11	Held Volume	$V_{hold} = A \cdot L_f \cdot \eta_v$	Held volume / حجم المادة داخل اللولب
12	Held Mass	$m_{hold} = \rho_b \cdot V_{hold}$	Held mass / كتلة داخل اللولب
13	Disc Volume	$V_{disc} = A_{disc} \cdot h$	Disc volume / حجم المادة فوق الصحن
14	Disc Mass	$m_{disc} = \rho_b \cdot V_{disc}$	Disc mass / كتلة المادة فوق الصحن
15	Weight Force	$F = m \cdot g$	Weight force / قوة الوزن

Symbols

D: Outer diameter | D_s: Shaft diameter

A: Annular area | p: Pitch

η_v : Volumetric efficiency

n: Number of turns

N: rev/s

ρ_b : Bulk density

ρ_b

m_{dose}: Target mass

-

θ

L_f: Flighted length

ρ_b

A_{disc}: Disc area

ρ_b

g: 9.81 m/s²

Engineering Notes

Base volumetric area | أساس كل الحسابات الحجمية

Ideal filling | يفترض امتلاءً كاملاً

Includes voids | يأخذ الفراغات بعين الاعتبار

Dose basis | يستخدم في حساب الجرعة

Speed dependent | مرتبط بـ سرعة الدوران

Torque input | مهم للتعبة الوزنية

Fill time | يستخدم لضبط زمن التعبئة

Control timing | أساس التحكم في السيرفو

Position control | يستخدم في التحكم الموضعي

Servo command | Position Control مهمة في

Axial load | للتأكد من عدم الانقطاع

Axial force | تدخل في حساب العزم

Disc load | لتحليل الأحمال

Axial load | تؤثر على العزم المحوري

Static load | تستخدم في التحليل الإنشائي

Explanation of the equation and its use

حساب المساحة الفعالة التي تشغلها المادة بين الريشة والعمود
حجم المادة النظرية المنقولة في لفة واحدة
تعديل الحجم النظري حسب كفاءة التعبئة
الحجم الذي يحتويه اللولب عبر جميع اللفات
حجم المادة المنقولة في وحدة الزمن
تحويل الحجم إلى كتلة باستخدام الكثافة الظاهرية
معدل الكتلة المعبأة في الزمن
الزمن اللازم لتعبئة كتلة محددة
عدد لفات اللولب لتعبئة الجرعة
زاوية الدوران المطلوبة لكل تعبئة
حجم المادة داخل الجزء الحلزوني
كتلة المادة الموجودة داخل اللولب
حجم المادة المحمولة على الصحن
وزن المادة المؤثرة على الصحن
القوة العمودية الناتجة عن المادة

البند

Part Number
Product
Product Name
Series
Type
Protection Class
Enclosure
Flange Square Dimension
Motor Lead-out Configuration
Motor Encoder Connector
Power Supply Capacity
Voltage Specifications
Rated Output
Rated Current (rms)
Holding Brake
Mass
Oil Seal
Shaft Type
Rated Torque
Continuous Stall Torque
Momentary Max. Peak Torque
Max. Current (o-p)
Regenerative Brake Frequency
Rated Rotational Speed
Max. Rotational Speed
Moment of Inertia of Rotor
Recommended Load-to-Rotor Inertia Ratio
Rotary Encoder
Encoder Resolution
Radial Load (Assembly)
Thrust Load A-direction (Assembly)
Thrust Load B-direction (Assembly)
Radial Load (Operation)
Thrust Load A,B-direction (Operation)
Ambient Temperature
Applicable Drivers
Driver Supply Voltage
Driver Interface
Frame Size
Dimensions
Key-way Dimensions
Mounting Screw
Warranty Period
Export Control
Intended Use

Safety Notes

MSMF042L1U1
 Servo Motor
 MINAS A6 Family Servo Motor
 MSMF Series
 Low Inertia
 IP67

محمي باستثناء جزء الدوران من عمود الإخراج وموصلات الموتور والإنكودر

60 mm
 Connector
 Connector
 0.9 kVA
 200V
 400W
 2.4 A
 Without
 1.2 kg
 With
 Key-way, center tap

1.27 N·m
 1.27 N·m
 3.82 N·m
 10.2 A
 Without option: No limit With option: No limit External Regenerative Resistor Part No.: DVoP4283
 3000 r/min
 6000 r/min
 $0.27 \times 10^{-4} \text{ kg} \cdot \text{m}^2$
 30 times or less
 23-bit Absolute/Incremental system

8,388,608

392 N
 147 N
 196 N
 245 N
 98.0 N
 Derating curve available (see page 4 of datasheet)
 MBDLN25NE, MBDLN25SE, MBDLN25SG, MBDLT25NF, MBDLT25SF
 Single/3-phase 200V
 RTEx, Pulse, Modbus (RS485/RS232), Analog
 B-Frame

60 mm flange, 129 mm length (installation), 76 mm height, 30 mm shaft length
 5h9 width, M5 depth 10
 M5, torque: 2.7–3.3 N·m
 1 year from purchase or 18 months from manufacture
 May require license if used in military applications
 General-purpose industrial equipment only

Must be installed by qualified personnel; fail-safe design recommended

البند

Part Number
Product
Series
Type
Control Method
System Support
Safety Function
Frame Size
Frequency Response
Supply Voltage
Input Power - Main Circuit
Input Power - Control Circuit
Encoder Feedback
Absolute System Support
Pulse Input
Pulse Output
Control Signal Input
Control Signal Output
Analog Output
Communication
Regenerative Resistor
Control Modes
Dimensions (W×H×D)
Mass
Mounting Type
Mounting Screw
Connector Types
Warranty Period
Export Control
Intended Use
Safety Notes
Environmental Conditions
Torque Screw Recommendation
Smoke Warning
Sulfur Warning
Overvoltage Warning
User Responsibility
Warranty Exclusions

MBDLN25SE

Servo Driver

A6SE Series

Position Control Type

Position Control Only

Incremental System Only (لا يدعم النظام المطلق)

Without Safety Function

B-Frame

3.2 kHz

Single/3-phase 200V

Single/3-phase 200 to 240V +10% -15% 50/60 Hz

Single-phase 200 to 240V +10% -15% 50/60 Hz

23-bit Absolute/Incremental (8388608 resolution)

✗ Pr.0.15 = 1 يجب ضبط) لا يدعم النظام المطلق

2 inputs (Photo-coupler input, Line receiver input)

4 outputs (Line driver: 3 outputs, Open collector: 1 output)

10 general-purpose inputs (قابلة للبرمجة)

6 general-purpose outputs (قابلة للبرمجة)

2 outputs (Analog monitor)

USB (للإعداد والمراقبة فقط)

لا يحتوي على مقاومة تجديدية داخلية (خارجية فقط)

3 modes: 1. Position control 2. Internal velocity command 3. Position + Internal velocity

55 × 150 × 130 mm

1.0 kg

Base mount / Rack mount (optional brackets)

2-M4 × 6 flat head screw

XA: Main & control power input XB: Motor & regenerative resistor X1: USB X2: RS232/485 X3: Safety function (يستخدم)

1 year from purchase or 18 months from manufacture

قد يتطلب تصريحًا إذا استخدم في تطبيقات عسكرية

General-purpose industrial equipment only

يجب التركيب من قبل فني مختص، تصميم آمن ضد الأعطال، لا يستخدم في تطبيقات حساسة (نووي، طبي، طيران...) إلا بعد استشارة الشركة
يُراجع الدليل للتفاصيل

M5 لمسامير 2.7–3.3 N·m

قد ينبعث دخان في حالة العطل (ما يعادل سيجارة واحدة)

لا يستخدم في بيئات تحتوي على كبريت أو غازات حامضية

لا يتجاوز الجهد المقنن حتى لا يؤدي إلى حريق أو تلف

المستخدم مسؤول عن التوافق الميكانيكي والكهربائي والقانوني

لا يشمل: 1. الاستخدام الخاطئ أو التعديل 2. التلف أثناء النقل أو السقوط 3. عدم الالتزام بالمواصفات 4. الكوارث الطبيعية 5. دخول مواد غريبة (

X4: InterfaceX5: External scale (غير مستخدم)X6: Encoder (غير مستخدم)

	البند	المواصفة
Part Number	AB060A-020-S2-P1	
Series	AB Series (High Precision Planetary Gearbox)	
Frame Size	AB060A	
Stage	2 Stage	
Ratio		20
Shaft Option	S2: Output Shaft with Key	
Backlash Class	P1: Reduced Backlash (≤ 5 arcmin)	
Nominal Output Torque (T2N)	60 Nm	
Emergency Stop Torque (T2NOT)	180 Nm ($3 \times T2N$)	
Max. Input Speed (n1B)	10,000 rpm	
Nominal Input Speed (n1N)	5,000 rpm	
Efficiency	$\geq 94\%$ (2-stage)	
Backlash (P1)	≤ 5 arcmin	
Torsional Rigidity	7 Nm/arcmin	
Max. Radial Load (F2rB)	3,250 N	
Max. Axial Load (F2aB)	765 N	
Service Life	10,000 hrs (continuous operation)	
Operating Temperature	-10°C to +90°C	
Lubrication	Synthetic oil (مدى الحياة)	
Protection Class	IP65	
Mounting Position	All directions	
Noise Level (at 3000 rpm, no load)	≤ 58 dB(A)	
Mass	1.9 kg	
Inertia (J1)	0.47 kg·cm ²	
Motor Shaft Diameter (C3)	14 mm (inferred from image and bushing note)	

البند	القيمة (mm)
D1	70
D2	5.5
D3	16
D4	50
D5	45
D6	M5 x 0.8P
D7	80
L1	60
L2	37
L3	7
L4	1.5
L5	25
L6	2
L7	6
L8	98
L9	4.8
L10	12.5
C1	70
C2	M5 x 0.8P
C4	34
C5	50
C6	8
C7	60
C8	19
C9	154
C10	13.5
B1	5
H1	18

Item	Symbol / Equation	Numerical Substitution
Filled Mass	m	0.5
Gravity Acceleration	g	-
Material Weight	$W = m \cdot g$	0.5×9.81
Outer Radius	r_o	$60/2$
Inner Radius	r_i	$16/2$
Mean Radius	$r_m = (r_o + r_i)/2$	$(0.03 + 0.008)/2$
Torque due to Weight	$T_{weight} = W \cdot r_m$	4.905×0.019
Friction Coefficient	μ	-
Axial Friction Force	$F_f = \mu \cdot W$	0.3×4.905
Friction Torque	$T_f = F_f \cdot r_m$	1.47×0.019
Wall Friction Torque	T_{wall}	Conservative
Bearing Friction Torque	$T_{bearing}$	Estimate
Inertia Torque	$T_{inertia}$	Startup estimate
Total Raw Torque	ΣT	$\text{Sum}(7 \rightarrow 13)$
Safety Factor	SF	-
Design Torque	$T_{design} = SF \cdot \Sigma T$	2.5×0.151
Gear Ratio	i	-
Gearbox Efficiency	η	-
Required Servo Torque	$T_s = T_d / (i \cdot \eta)$	$0.378 / (20 \times 0.95)$
Servo Rated Torque	Datasheet	MSMF042L1U
Servo Peak Torque	Datasheet	-
Gearbox Output Torque	$T_{out} = T_s \cdot i \cdot \eta$	$0.16 \times 20 \times 0.95$
Effective Safety Factor	T_{out} / T_{design}	$3.04 / 0.378$
Design Status	-	-

Result	Unit	Engineering Note
0.5	kg	Target fill weight
9.81	m/s ²	Physical constant
4.905	N	Weight force
0.030	m	Auger outer radius
0.008	m	Shaft radius
0.019	m	Force application radius
0.093	N·m	Basic conveying torque
0.30	-	Fine salt
1.47	N	Material resistance
0.028	N·m	Internal friction torque
0.010	N·m	Flight-wall contact
0.0002	N·m	Bearing losses
0.020	N·m	During acceleration
0.151	N·m	Without safety factor
2.5	-	Industrial practice
0.378	N·m	Required on auger
20	-	Planetary gearbox
0.95	-	High precision gearbox
0.020	N·m	Actual servo load
0.16	N·m	Continuous operation
0.48	N·m	Acceleration peak
3.04	N·m	Available torque
≈8	-	Very safe
SAFE	-	No resizing required

No.	Torque Name	Equation
1	Material Weight Torque	$T_m = m \cdot g \cdot r_m$
2	Axial Friction Torque	$T_a = \mu \cdot m \cdot g \cdot r_m$
3	Shear Torque	$T_s = \tau \cdot A \cdot r_m$
4	Wall Friction Torque	$T_w = \mu_w \cdot N \cdot r_o$
5	Bearing Friction Torque	$T_b = f_b \cdot F_r \cdot r_b$
6	Coupling/Key Loss Torque	$T_k \approx \mu \cdot F \cdot r$
7	Screw Inertia Torque	$T_J = J \cdot \alpha$
8	Material Inertia Torque	$T_{Jm} = J_m \cdot \alpha$
9	Total Raw Torque	$T_{raw} = \sum T$
10	Design Torque	$T_{design} = SF \cdot T_{raw}$
11	Required Servo Torque	$T_{servo} = T_{design} / (G \cdot \eta)$
12	Gearbox Output Torque	$T_{out} = T_{servo} \cdot G \cdot \eta$
13	Actual Safety Factor	$SF_{actual} = T_{out} / T_{design}$

Symbols	Description	Engineering Purpose
m :	mass, g : gravity, r_m : mean radius	Torque due to material weight
μ :	friction coefficient	Axial resistance of material
τ :	shear stress, A : shear area	Material internal shear
μ_w :	wall friction, N : normal force	Flight-wall friction
f_b :	bearing factor	Bearing resistance
-		Losses in coupling/key
J :	inertia, α : angular accel.	Startup torque
J_m :	material inertia	Material dynamic torque
-		Sum of all torques
SF:	safety factor	Design condition
G :	gear ratio, η : efficiency	Motor sizing
-		Available torque
-		Safety check